Photoelectric Effect

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Photoelectric Effect

Introduction

Radiation and matter display both wave-like and particle-like properties. Energy is quantized into discrete packets called the Photon.

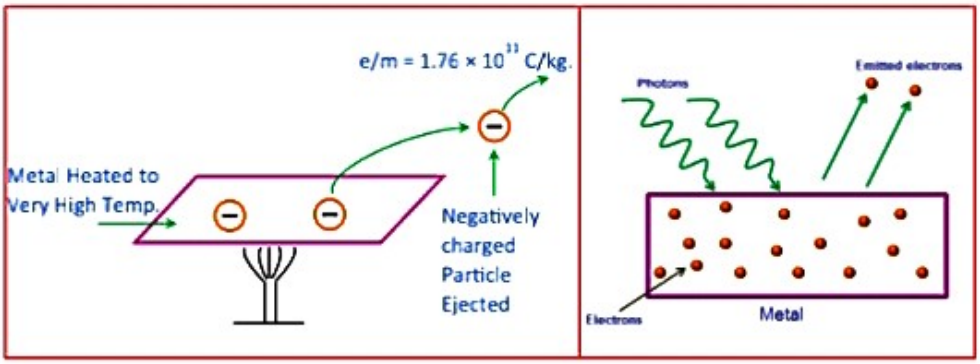
"In 1887, the maxwell's equations of electromagnetism and hertz experiments on the generation and detection of electromagnetic waves strongly established the wave nature of light."

The cathode rays were discovered in 1870 by William Crookes, who later, in 1879, suggested that these rays consisted of steams of fast-moving negatively charged particles. J.J. Thomson confirmed this hypothesis. J.J. Thomson was the first scientist to determine the speed ($0.3 \times 10^8 - 0.6 \times 10^8$ m/s) and the specific charge (charge to mass ratio) experimentally (e/m) of the cathode ray particles.

The presently accepted value of e/m is (1.76×10^{11} C/kg). Further, the value of e/m was found to be independent of the nature of the material/metal used as the cathode (emitter), or the gas introduced.

"The observation suggested the universality of the cathode ray particles."

In 1887, it was found that certain metals, when irradiated by ultraviolet light, emitted negatively charged particles having small speeds. Also, certain metals when heated to a high temperature were found to emit negatively charged particles. The value of e/m of these particles was found to be the same as that for cathode ray particles.



These observations thus established that all these particles, although produced under different conditions, were identical in nature. J.J Thomson, in 1897 named these particles as electrons and suggested that they were fundamental, universal constituents of matter.

Electron Emission

The emission of the electron from the metallic surface is defined as electron emission.
The electrons in metal that are free to move within the volume of metal are called free electrons.
These electrons do not get ejected out of the surface of metal on their own because of the potential barrier existing all around its surface. The minimum amount of work or energy necessary to take a free electron out of metal against the attractive forces of positive ions inside a metal is called the work function of that metal.

Different Methods for Electron Emission

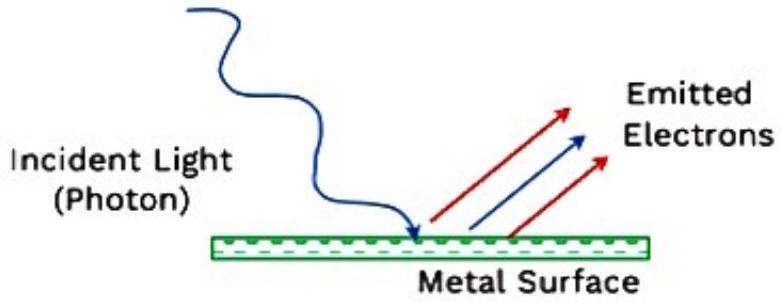
(1) **Photoelectric Effect:-** When electromagnetic radiations of suitable wavelength or frequency are incident on a metallic surface, then electrons are emitted; this phenomenon is called the photoelectric effect.

(2) **Thermionic Emission: -** In this case, extra energy is given to electrons to overcome the potential barrier in the form of heat by passing a current through a filament (joule's heating effect).

(3) **Field Emission: -** In this case, metal is placed in a strong electric field due to which the electrons are accelerated to such a speed that the corresponding kinetic energy is sufficient to overcome the potential barrier.

(4) **Secondary Emission: -** Electrons from metal can also be ejected by impinging a beam of high-speed electrons so that through collisions, electrons acquire sufficient Kinetic Energy to overcome the potential barrier. This way of Electron Emission is called secondary emission.

Photo-Electric-Effect

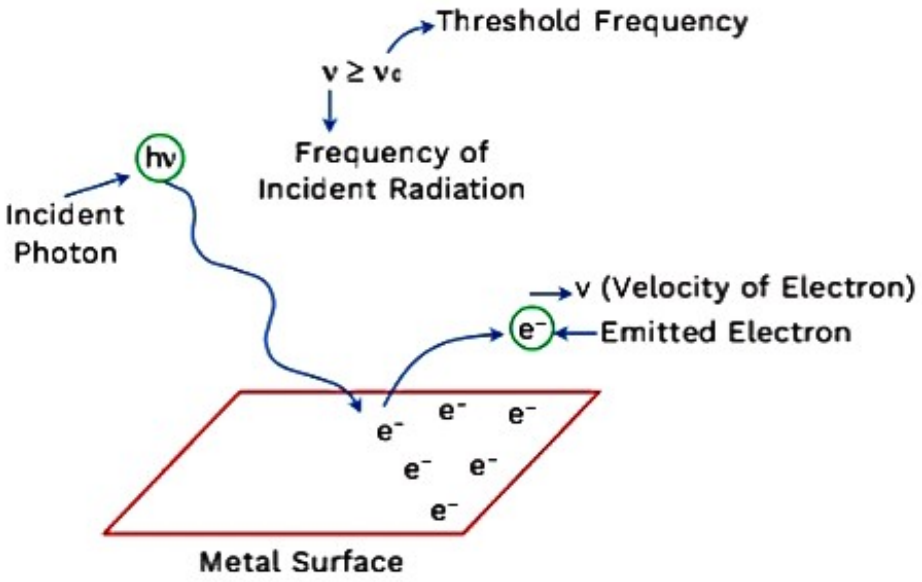


When electromagnetic radiation of suitable wavelength or frequency is incident on a metal surface due to this electrons are emitted, this phenomenon is called the photoelectric effect. The ejected electrons are called photoelectrons. Current flow due to the flow of photoelectrons is called photocurrent.

This effect was discovered by Hertz. Laws of photo electric effect were given by Lenard. It was explained by Einstein using the quantum theory of light.

Some Standard Definitions

(1) Threshold Frequency: The minimum frequency of incident radiation below which the photoelectric effect does not occur is known as the threshold frequency.



Then;

(i) If $\nu < \nu_0$, no ejection of photo electrons.

(ii) $\nu = \nu_0$, photo-electrons are ejected from metal surface and their K.E. is zero.

(iii) If $\nu > \nu_0$, then photo electrons will come out of the surface along with kinetic energy.

2. Threshold Wavelength (λ_0) : "The maximum wavelength above which photo electric effect does not occur is known as threshold wavelength."

$$\nu_0 \propto \frac{1}{\lambda_0}$$

$$\boxed{\nu_0 = \frac{c}{\lambda_0}}$$

Minimum frequency → Maximum Wavelength

$$\boxed{\lambda_0 = \frac{c}{\nu_0}}$$

3. Work Function or Threshold Energy : The minimum energy of incident photon below which no ejection of photoelectron from the metal surface will take place is known as work function or threshold energy for that metal.

$$\phi = h\nu_0 = \frac{hc}{\lambda_0(max)}$$

ν_0 = Threshold Frequency : ϕ = work function

λ_0 = Threshold wavelength

Important Point:-

Work function is the characteristics of a given metal.

	Metal	Work Function
1.	Celsius	1.9 eV
2.	K	2.2 eV
3.	Na	2.3 eV
4.	Li	2.5 eV
5.	Ca	3.2 eV
6.	Cu	4.5 eV
7.	Ag	5.6 eV

4. Stopping Potential :-

The magnitude of the minimum negative potential at which photoelectric current becomes zero is called stopping potential (V_0).

It means stopping potential measures the maximum kinetic energy of photo-electron.

$$K_{max} = eV_0$$

Maximum Kinetic Energy — Stopping Potential

Failure of Wave-Model of Light

(1) According to wave theory, when light is incident on a surface, energy is distributed continuously over the surface. So, the electrons must take a time interval to accumulate a sufficient amount of energy to come out from the surface. But in the experiment, there is no time lag observed.

(2) When intensity is increased, more energetic electrons should be emitted. So, the stopping potential should be intensity-dependent, but it is not observed.

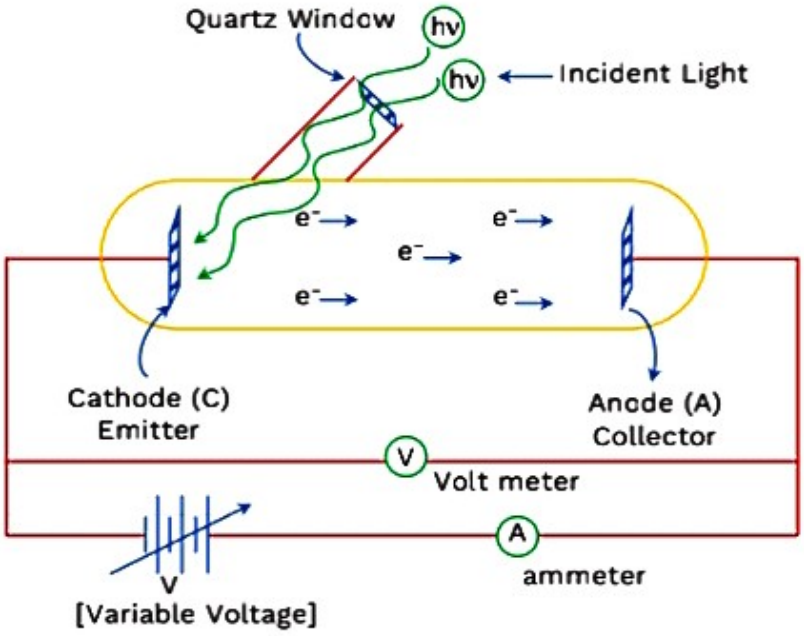
(3) According to wave theory, if the intensity is sufficient, then, at each frequency, electron emission is possible. It means there should not be the existence of threshold frequency.

Hallwach and Lenard's Observation :-

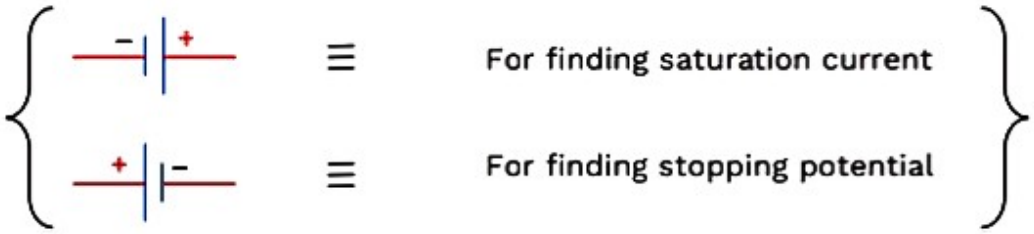
(1) **Hallwach Experiment:-** Hallwach observed that if negatively charged Zn plate of the electroscope is illuminated by ultraviolet light, its negative charge decreases and it becomes neutral, and after some time, it gains positive charge. It indicates that under the action of ultraviolet light, some negatively charged particles are emitted from the metal.

(2) **Lenard Experiment :-** He told that when ultraviolet rays are incident on the cathode, electrons are ejected, these electrons are attracted by the anode, and due to complete path of photoelectrons, photocurrent flows, when ultraviolet rays are incident on the anode, electrons are ejected, but current does not flow.

Experimental Study of Photo Electric Effect



The polarity of Variable Voltage



It consists of two conducting electrodes, the cathode (C), also called an emitter, and anode (A), also called a collector, which is enclosed in an evacuated glass tube, as shown in the figure.

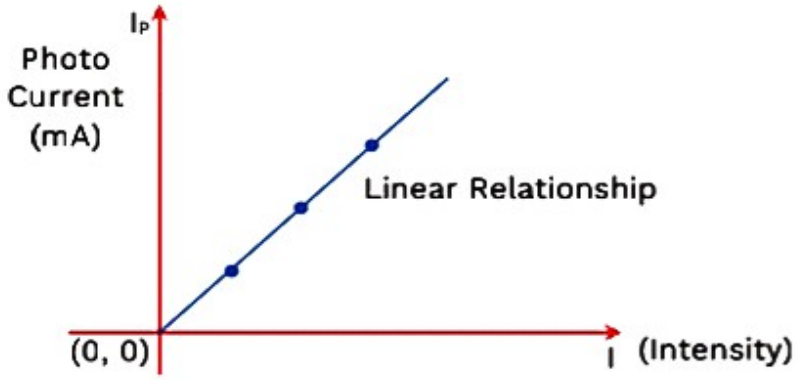
Light of a certain wavelength or frequency falling on the surface of the cathode causes a current to flow in the external circuit. This current is called the 'photoelectric current' (i_p).

Now we are going to study about four effects: -

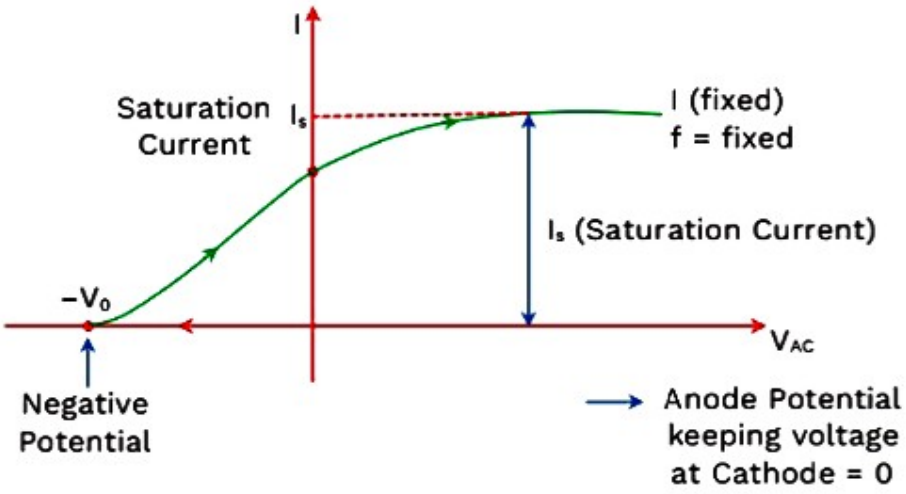
(1) Effect of intensity.
(2) Effect of Potential difference between Anode (A) and Cathode (C).
(3) Effect of frequency of incident Radiation.
(4) Time Lag.

(1) Effect of Intensity: - Keeping potential of collector fixed note the value of current in the circuit for different intensity of light.

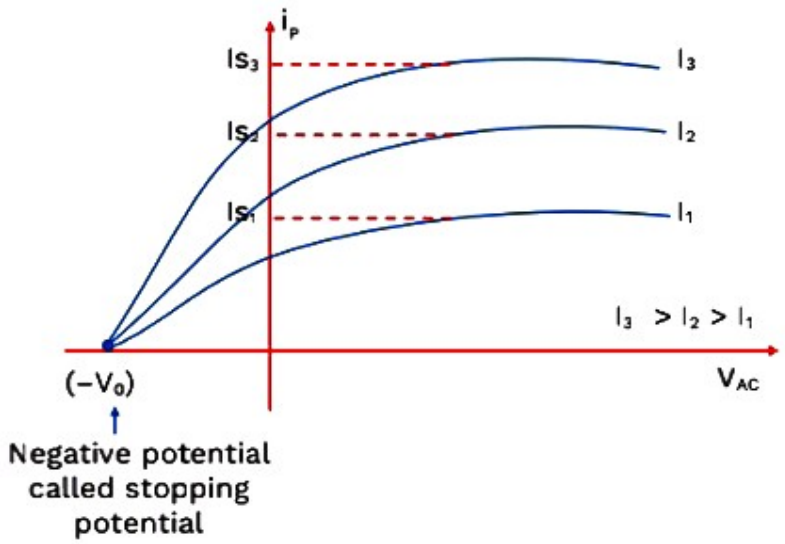
"The photoelectric current is directly proportional to the intensity of incident radiation."



(2) The Effect of Potential Difference Between Anode and Cathode (A & C) :-
Light of fixed frequency (ν) and intensity (I) is incident on photo emissive plate 'C,' keeping 'A' positive with respect to 'C.' If the positive potential of 'A' is gradually increased, it is found that the first photocurrent increases then becomes maximum and this maximum current is called 'Saturation Current.'



Now the above graph shown at different intensities



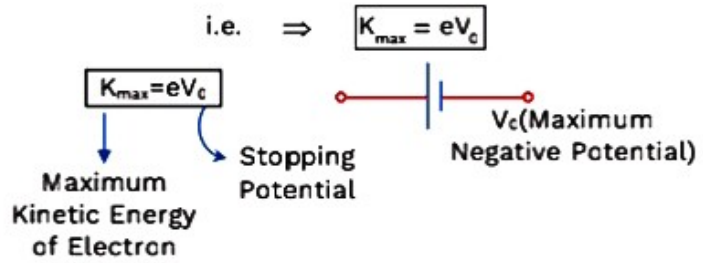
Important Point :-

(1) Frequency (f) is fixed, so stopping potential is fixed ($-V_0$).

(2) If frequency (f) is fixed and if we increase the Intensity, the saturation current increases."

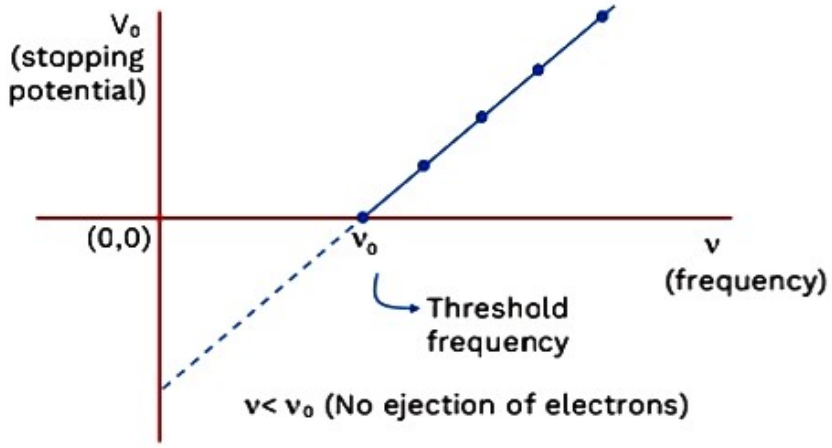
Stopping Potential :-

The magnitude of the minimum negative potential at which i_p becomes zero is called stopping potential (V_0). If means stopping potential measures maximum kinetic energy of photoelectrons.



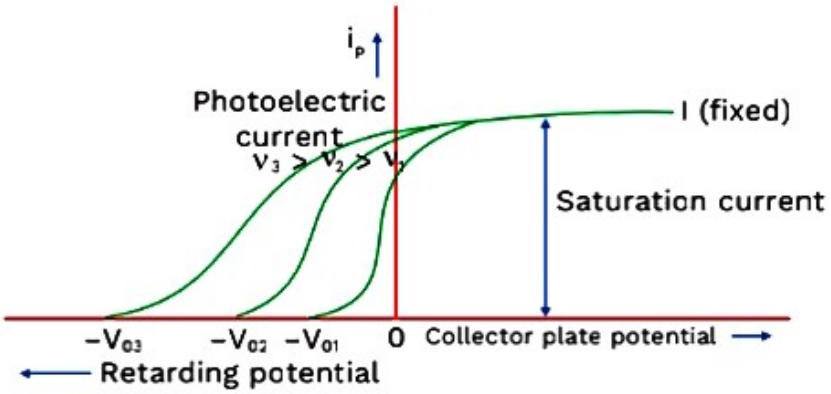
(3) Effect of Frequency: -

Stopping potential V_0 varies linearly, with a frequency of incident radiation, and it is more negative at a higher frequency.



Important Point :-

Keeping the intensity of incident radiation constant, the variation of potential difference and current is obtained as shown in the graph :



(4) Time Lag : There is no time lag between the incidence of radiations and the emission of electrons. It is a spontaneous process.

Explanation of Photo Electric Effect by Einstein (Einstein's Photo Electric Equation) :-

In 1905, Albert Einstein proposed a radically new picture of electromagnetic radiation to explain the electric photo effect.

There are the following points explained by Einstein :-

(1) Radiations absorbed by the surface area in the form of quanta (Photon). The energy of each Photon depends on frequency. One Photon can interact with one electron at a time.

In the interaction between Photon and electron, the incident photon transfers its whole energy to the electron. If energy is sufficient, then the electron comes out without any time delay. It means the electric photo effect is an instantaneous process

(2) If the intensity of the given source is increased, then the number of photons increases so that a greater number of electrons are emitted, and a greater saturation current is obtained. It means saturation current depends upon the intensity of the given source.

$$\boxed{i_s \propto I}$$

(3) At a time, only one Photon can interact with one electron, and during the interaction, the electron uses the energy of the Photon.

By energy conservation law,

Energy of photon ($h\nu$) = kinetic energy of electrons + Energy required to make electron free from metal + energy lost

$h\nu =$ KE $+$ ϕ_0 $+$ Q

- KE → Kinetic Energy of Electron
- ϕ_0 → Work Function
- Q → Energy Lost in Collision

If Q = 0 (In a collision, energy lost is zero), then the kinetic energy of the electron is maximum

Now,

$$\boxed{h\nu = (K.E)_{max} + \phi_0} \quad \ldots(1) \rightarrow$$ Known as Einstein's equation of Photo Electric Effect.

If V_0 is stopping potential :-

$(KE)_{max} = eV_0$

$h\nu = eV_0 + \phi_0$

$eV_0 = h\nu - \phi_0$

$$\boxed{V_0 = \left(\frac{h}{e}\right)(\nu) - \frac{\phi_0}{e}} \quad \text{... (2)}$$

"It means maximum kinetic energy and stopping potential depend on frequency. It is independent of the intensity of the given source."

Work function $(\phi_0) = h\nu_0$

$h\nu = (KE)_{max} + \phi_0$

$h\nu = (KE)_{max} + h\nu_0$

$h\nu = eV_0 + h\nu_0$

$eV_0 = h\nu - h\nu_0$

$$V_0 = \left(\frac{h}{e}\right)(\nu) - \frac{h\nu_0}{e}$$

$$\boxed{V_0 = \left(\frac{h}{e}\right)(\nu) - \left(\frac{h}{e}\right)(\nu_0)} \quad \text{...(3)}$$

V_0 = Stopping Potential.
h = Plank Constant = 6.626×10^{-34} Jr
e = Electronic Charge.
ν_0 = Threshold frequency.

Graphs :-

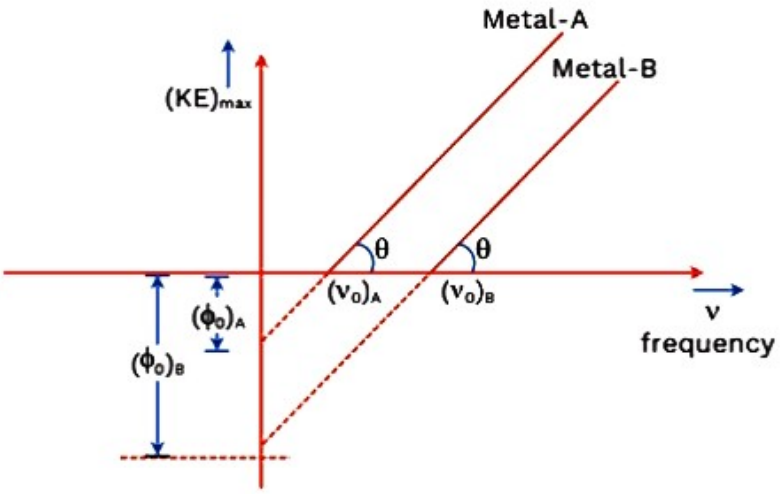
(1) Graph between $(KE)_{max}$ and frequency

$(KE)_{max} = h\nu - \phi_0$

Compare; $y = mx + C$

Slope (m) = $\tan\theta = h$ (Same for all metals)

$(\phi_0)_B > (\phi_0)_A$

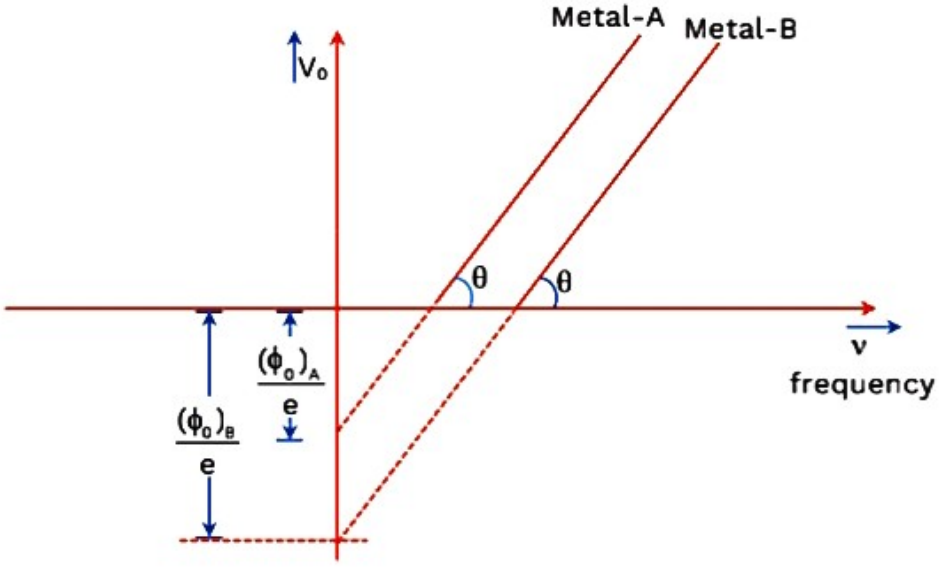


(2) Graph between stopping potential (V_0) and frequency (ν)

$$eV_0 = h\nu - \phi_0$$

$$V_0 = \left[\frac{h}{e}\right](\nu) - \left[\frac{\phi_0}{e}\right]$$

Slope = m = $\tan\theta = \frac{h}{e}$ (same for all metals)



Einstein's Theory of Photons:-

The photoelectric effect thus gave evidence to the strange fact that light in interaction with matter behaved as if it was made of quanta or packets of energy, each of energy hν.

Einstein arrived at the important result, that the light quantum can also be associated with momentum $\left(\frac{h\nu}{c}\right)$. A definite value of energy, as well as momentum, is a strong sign that the light quantum can be associated with a particle. This particle was later named as photon.

Important Points About Photon: -

(1) In the interaction of radiation with matter, radiation behaves as if it is made up of particles called photons.

(2) Each photon has energy ($E = h\nu$) and momentum $\left(p = \frac{h\nu}{c}\right)$, and speed 'c', the speed of light.

(3) All photons of light of a particular frequency ν, or wavelength λ, have the same energy $\left(E = h\nu = \frac{hc}{\lambda}\right)$ and momentum $\left(p = \frac{h\nu}{c} = \frac{h}{\lambda}\right)$, whatever the intensity of the radiation may be. By increasing the intensity of light of a given wavelength, there is only an increase in the number of photons per second crossing a given area, with each photon having the same energy. Thus, photon energy is independent of the intensity of radiation.

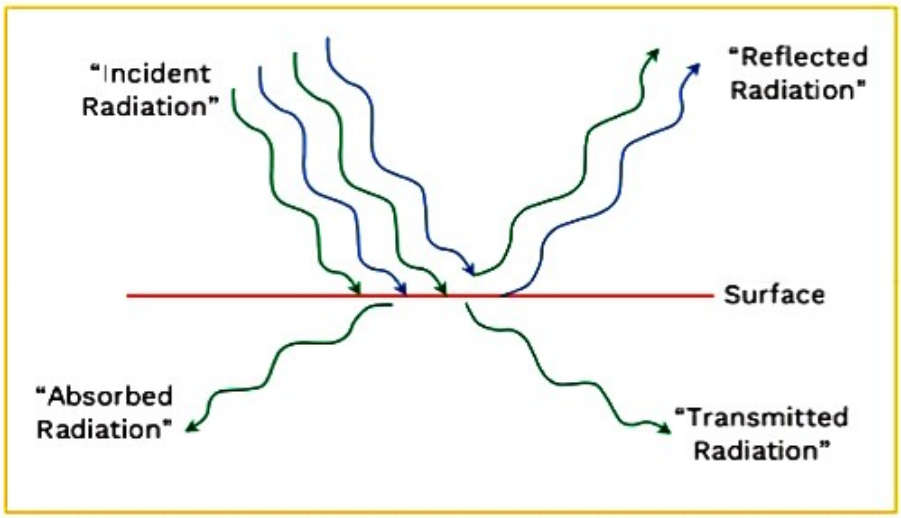
(4) Photons are electrically neutral and are not deflected by electric and magnetic fields.

(5) In a photon-particles collision (such as photon-electron collision), the total energy and total momentum are conserved. However, the number of photons may not be conserved in a collision. The photon may be absorbed, or a new photon may be created.

Force and Radiation Pressure

We know that each photon has a definite energy and a definite linear momentum. Every photon in the light beam of a particular wavelength 'λ' has the same energy, i.e., $\frac{hc}{\lambda}$ and the same momentum $\frac{h}{\lambda}$.

When the light of intensity falls on a surface, it exerts a force on that surface. This is because the incident light may be reflected (Partially or completely), due to which there will be a change in its momentum, and hence a force is exerted on the surface.



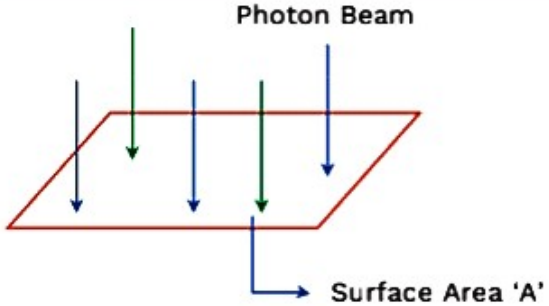
Assumption

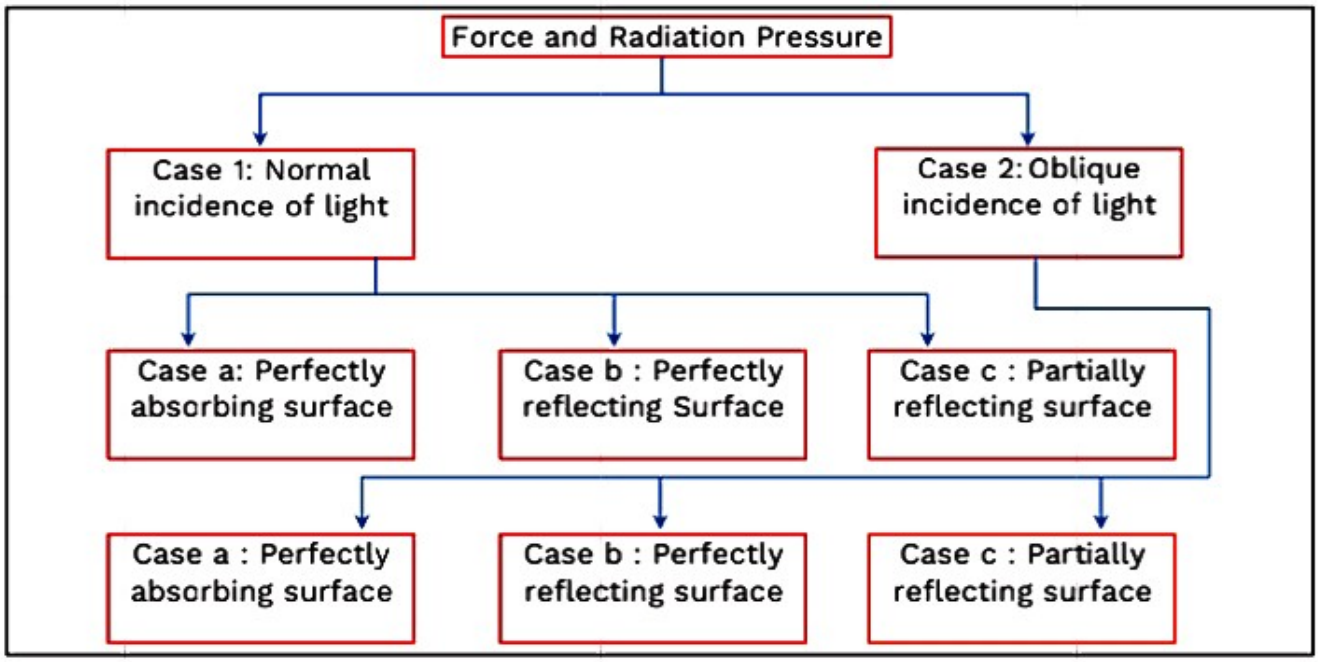
Assume Transmitted Radiation energy = 0

$$\text{Absorptive coefficient } (a) = \frac{\text{Energy Absorbed}}{\text{Energy Incident}}$$

$$\text{Reflective coefficient } (r) = \frac{\text{Energy Reflected}}{\text{Energy Incident}}$$

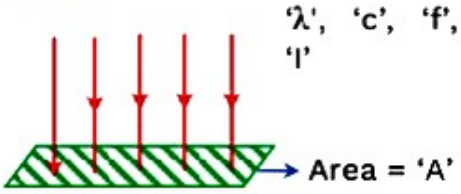
Let us assume that light beam of intensity I, wavelength λ, energy E, having N photons, be incident normally on the surface having area 'A' as shown in the figure.

Energy of each photon $E = \frac{hc}{\lambda}$ Momentum of each photon $P = \frac{h}{\lambda}$	



Case 1 :- Force and radiation pressure due to a photon beam incident normally on a surface.

SubCase (a): r = 0 ; a = 1



The initial Momentum of each photon $P = \frac{h}{\lambda}$

Since the photon gets completely absorbed by the surface, so final momentum of the photon is $p_f = 0$

Taking the direction of incidence of photon as positive, the change in momentum of each incident photon is

$$\Delta p_{\text{each photon}} = \left|0 - \frac{h}{\lambda}\right| = \frac{h}{\lambda}$$ Upward direction

So, the total change in momentum of the incident photon beam having N photon is -

$$\Delta p = N \times (\Delta p \text{ each phton})$$

$$(\Delta p)_{\text{Total momentum change}} = N \times \Delta p$$

$$\text{Intensity} = \frac{\text{Energy}}{\text{Area} \times \text{Time}}$$

$$I = \frac{\text{Energy}}{A \times t}$$

Energy = $I \times A \times t$

The energy of one photon $= \frac{hc}{\lambda}$

If N is total no of photon

Then total energy $= N \times \frac{hc}{\lambda} = I \times A \times t$

Number of photon incident in time 't' $\longrightarrow$ $\boxed{N = \frac{I \times A \times t}{\left(\frac{hc}{\lambda}\right)}}$

$$(\Delta p)_{Total} = N \times \left(\frac{h}{\lambda}\right)$$

$$(\Delta p)_{Total} = \frac{I \times A \times t}{\frac{hc}{\lambda}} \times \frac{h}{\lambda}$$

$$= \frac{I \times A \times t}{hc} \times \lambda \times \frac{h}{\lambda}$$

$$(\Delta p)_{Total} = \frac{I \times A \times t}{c}$$

$$\frac{(\Delta p)_{Total}}{t} = \frac{I \times A}{c}$$

$$\boxed{\text{Force exerted on surface } (F_{exerted}) = \frac{I \times A}{c}}$$

Now,

The pressure exerted on surface:-

$$\frac{Force}{A} = \frac{I}{c}$$

$$\boxed{\text{Radiation Pressure, } P = \frac{I}{c}}$$

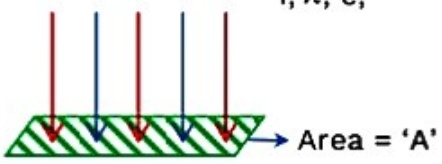
Where,

P = Radiation Pressure
I = Intensity of light
c = Velocity of light
A = Area of cross-section
F = Force exerted on surface

ub Case b : a = 0 (No Absorption)
r = 1 (Complete reflection)

Given Parameter :

I, λ, c,



$p_i = \frac{h}{\lambda}$ (downward)

$p_f = \frac{h}{\lambda}$ (upward)

$\Delta p = p_f - p_i$

Let downward direction taken a positive direction.

$$(\Delta p) = -\frac{h}{\lambda} - \left(\frac{h}{\lambda}\right)$$

$$(\Delta p) = -\frac{2h}{\lambda}$$

$$|\Delta p| = \left|-\frac{2h}{\lambda}\right|$$ (Upward direction since sign come out to be negative)

$$\boxed{|\Delta p| = \frac{2h}{\lambda}}$$

$$(\Delta p)_{Total} = N \times \frac{2h}{\lambda}$$

$N = \frac{I \times A \times t}{\left(\frac{hc}{\lambda}\right)}$ ...(As discussed in previous article)

$$(\Delta P)_{Total} = \frac{I \times A \times t}{\frac{hc}{\lambda}} \times \frac{2h}{\lambda}$$

$$(\Delta P)_{Total} = \frac{2I \times A \times t}{c}$$

$$\frac{(\Delta P)\,Total}{t} = \frac{2I \times A}{c}$$

$$\boxed{\text{Force on surface} = \frac{2IA}{c}}$$

$$\boxed{\text{Radiation Pressure } P = \frac{F}{A} = \frac{2I}{c}}$$

Sub Case C: r = r (Reflection coefficient)

a = 1 – r (Absorption coefficient)

Force
- Force due to absorption
- Force due to reflection

F
- $a \times \frac{(\Delta p)_{Total}}{t}$ Due to absorption
- $r \times \frac{(\Delta p)_{Total}}{t}$ Due to reflection

$$\text{Due to absorption} = (a) \times \frac{(\Delta p)_{Total}}{t}$$

$$= (a) \times \frac{I \times A}{c} \quad \text{...(As discussed in previous article)}$$

$$\text{Due to reflection} = r \times \left(\frac{2I \times A}{c}\right) \quad \text{...(As discussed in previous article)}$$

$$F_{Total} = (a)\left(\frac{I \times A}{c}\right) + r \times \left(\frac{2I \times A}{c}\right)$$

$$= (1 - r)\frac{I \times A}{c} + r \times \left(\frac{2I \times A}{c}\right)$$

$$F_{Total} = \frac{I \times A}{c}[1 + r]$$

$$\boxed{F = \frac{I \times A}{c}[1 + r]}$$

$$\frac{F}{A} = \frac{I}{c}[1 + r]$$

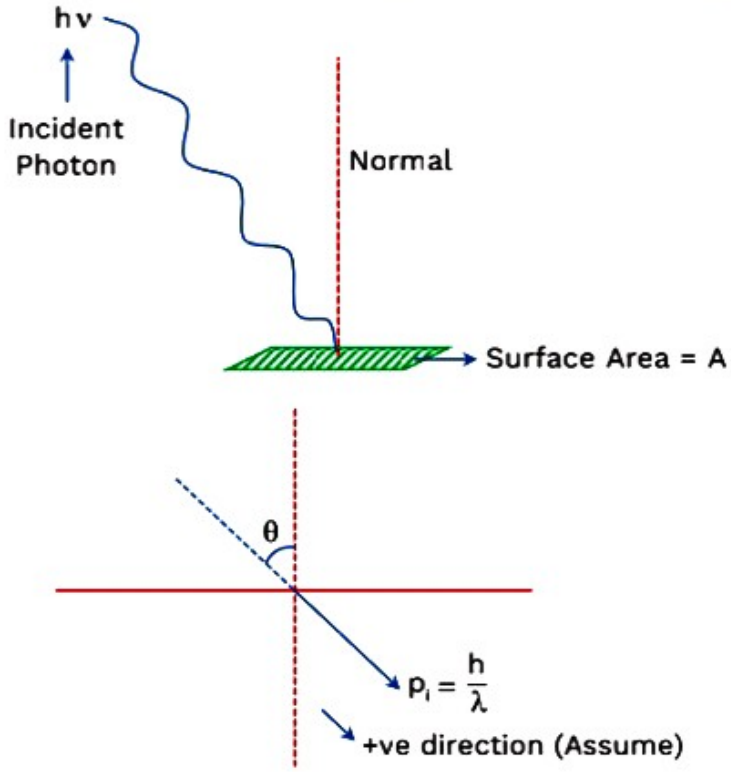
$$\boxed{P_{\text{Radiation Pressure}} = \frac{I}{c}(1 + r)}$$

Case 2 :- Force and Radiation Pressure Due to Oblique incidence of Photon Beam:

Subcase a: a = 1 (Complete Absorption) ; r = 0 (No reflection)

Let a light beam of intensity I, wavelength λ, energy E, power 'P', having N – photons, be

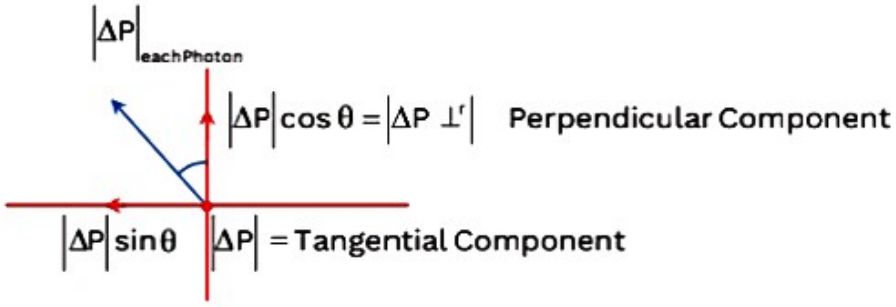
incident on the surface having area 'A' at an angle θ as shown in the figure.



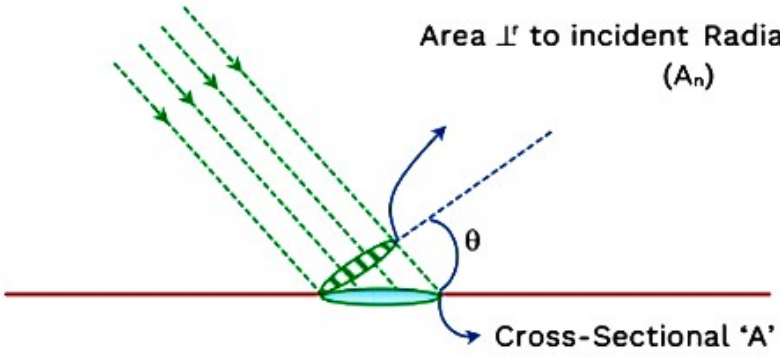
Since, (r = 0) so no photon is reflected ($p_r = 0$).

Change in momentum (Δp) for each photon $= \vec{p_f} - \vec{p_i}$

$\left|\left(\overrightarrow{\Delta p}_{each}\right)\right| = \left(0 - \frac{h\nu}{\lambda}\right)$; direction is making angle θ with normal.



Now according to the definition of intensity, "The Intensity I of the light beam is defined as the energy incident per second per unit area of a surface held normally to the direction of propagation of light".



$$I = \frac{\text{Energy}}{\text{Normal Area} \times t}$$

Energy = $I \times (A \cos\theta) \times t$

Energy of one Photon $= \frac{hc}{\lambda}$

Number of Photon = N

$$N \times \frac{hc}{\lambda} = I \times (A\cos\theta) \times t$$

$$N = \frac{I \times (A\cos\theta) \times t \times \lambda}{hc} \qquad \ldots(2)$$

Now Total momentum change perpendicular to the surface

$$\left|\overline{\Delta p}\right| = N \times \left|\overline{\Delta p}\right|_{\text{each Photon in } \perp^r \text{ direction}}.$$

$$= \frac{I \times A\cos\theta \times t \times \lambda}{hc} \times \left|\Delta P\right| \cos\theta$$

$$= \frac{I \times A\cos\theta \times t \times \lambda}{hc} \times \frac{h}{\lambda} \times \cos\theta$$

$$\left|\overline{\Delta P}\right|_{\perp^r} = \frac{IA\cos^2\theta \times t}{c}$$

$$\frac{\left|\overline{\Delta P}\right|_{\perp^r}}{t} = \boxed{\vec{F}_{\perp^r} = \left(\frac{I\cos^2\theta}{c}\right) A}$$

Component of force acting ⊥r to the Surface Area.

Now according to the definition of Pressure $= \frac{\vec{F}_{\perp'}}{\text{Area}}$

$$\boxed{\text{Radiation pressure} = \frac{\vec{F}_{\perp'}}{A} = \frac{I\cos^2\theta}{c}}$$

Sub Case –b :

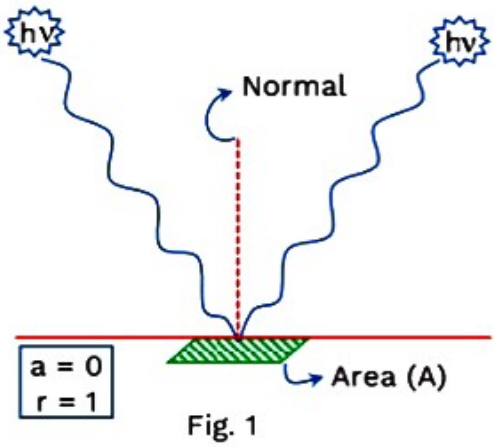


Fig. 1

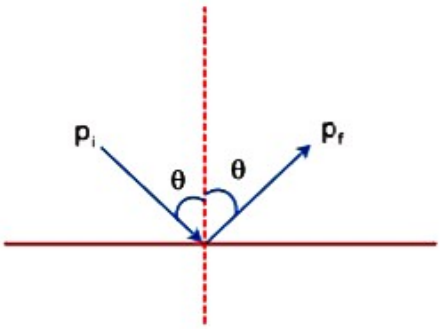


Fig. 2

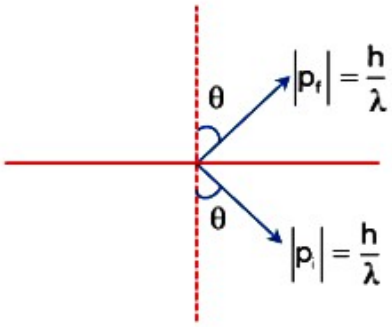


Fig. 3

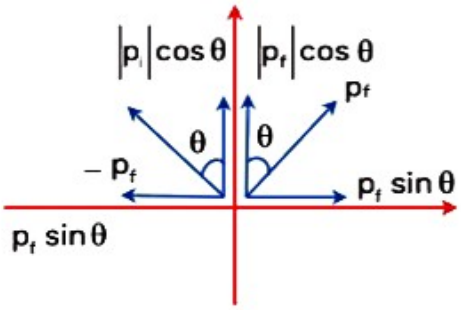


Fig. 4

$\left|\overrightarrow{\Delta p}\right|_{\perp}$ = change in momentum for one photon perpendicular to direction of surface

$$\left|\overrightarrow{\Delta p}\right|_{\text{Normal Component for one Photon}} = \vec{p}_f - \vec{p}_i = 2 \times \left|\frac{h}{\lambda}\right| \cos\theta$$

$$\boxed{\left|\overrightarrow{p_{f\perp'}}\right| = \frac{2h}{\lambda}\cos\theta}$$

$$\left|\overrightarrow{\Delta p}\right|_{\text{Total } \perp' \text{ to the surface}} = N \times 2\frac{h}{\lambda}\cos\theta$$

$$N = \frac{I(A\cos\theta)\times t}{\left(\frac{hc}{\lambda}\right)}$$ (Calculation in previous article)

$$\left|\overrightarrow{\Delta p}\right|_{Total} = \frac{I(A\cos\theta)t}{\frac{hc}{\lambda}} \times \left(2\frac{h}{\lambda}\cos\theta\right)$$

$$= \frac{2IA\cos^2\theta \times t}{c}$$

$$\frac{\left|\overrightarrow{\Delta p}\right|_{Total}}{t} = \frac{2IA\cos^2\theta}{c}$$

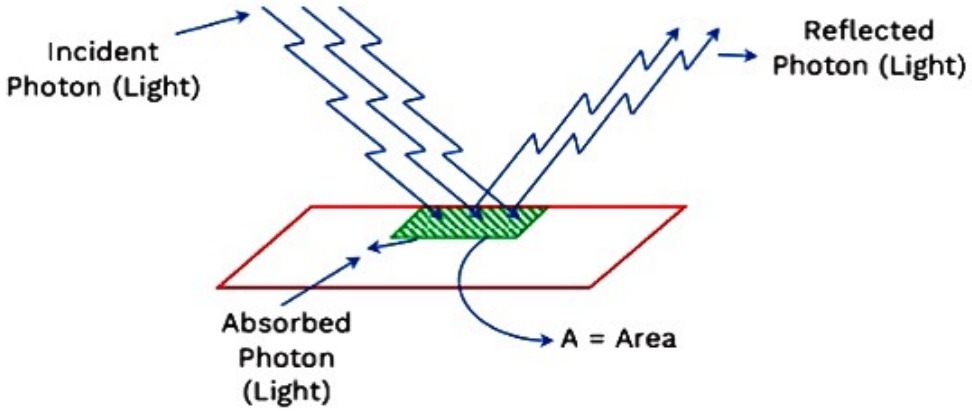
$$\boxed{\left|\vec{F}_{\perp'}\right| = \frac{2IA\cos^2\theta}{c}}$$

Radiation Pressure $\frac{\left|\vec{F}_{\perp'}\right|}{A} = \frac{2I\cos^2\theta}{c}$

$$\boxed{P = \frac{2I\cos^2\theta}{c}}$$

Case c. Reflection Coefficient = r

Absorption Coefficient = a = (1−r)



Radiation Pressure due to absorbed photon

$$= \left[\frac{I}{c}\cos^2\theta\right][a] = \left(\frac{I}{c}\cos^2\theta\right)(1-r)$$

↓

{Concept discussed in previous Article}

Radiation Pressure due to reflected photon

$$= \left[\frac{2I}{c}\cos^2\theta\right][r]$$

↓

{Concept discussed in the previous Article}

So,

Net Radiation Pressure = Pressure due to absorbed Photon + Pressure due to reflected Photon.

$$P_{net} = P_{Ab} + P_{ref}$$

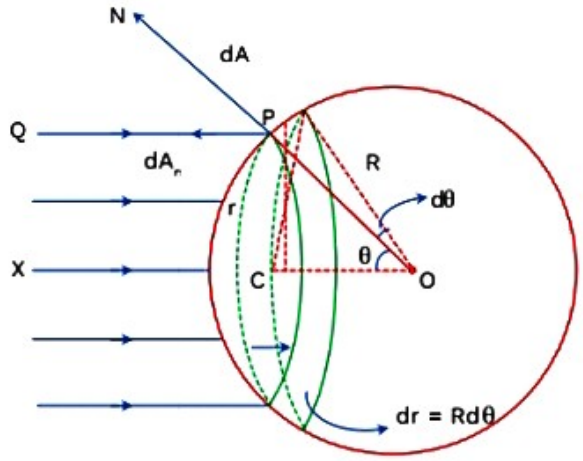
$$= \left[\frac{I}{c}\cos^2\theta\right][1-r] + \left[\frac{2I}{c}\cos^2\theta\right][r]$$

$$= \frac{I}{c}\cos^2\theta[1-r+2r]$$

$$\boxed{P_{Net} = (1+r)\frac{I}{c}\cos^2\theta}.$$

Q. **Given that a perfectly reflecting solid sphere of radius R is placed in the path of a uniform beam of large cross-sectional area and intensity, I. Calculate the force exerted on this sphere due to the light beam. Also, calculate the force on the sphere if it would have been perfectly absorbing. Support the result obtained by an analytical argument.**

Sol. Assume that O be the centre of the sphere and QP be the light incident on the sphere of Radius 'R'. Let I be the intensity of the incident light. Consider a circular element of radius r, thickness dr, subtending on angle $d\theta$ at the centre of the sphere as shown in the figure.



Then we observe that $r = R\sin\theta$ and $dr = rd\theta$. If dA be the area of this circular element, then

$$dA = (2\pi r)\, dr = 2\pi\,(R\sin\theta)\,(Rd\theta) = 2\pi R^2 \sin\theta d\theta$$

Since intensity I is defined as the energy incident per second per unit area of a surface held normal to the direction of propagation of light, hence;

$$I = \frac{dE}{(dA_n)\times(dt)}.$$

Where $dA_n = dA\cos\theta$ is the Area of the element normal to the direction of propagation of light.

$$\Rightarrow \qquad I = \frac{dE}{(dA\cos\theta)(dt)}$$

So, the energy of the light falling on this element of area dA in time dt is

$$\boxed{dE = Idt(dA\cos\theta)}$$

The momentum (dp) of light on this element of area dA is $\frac{dE}{c}$ along QP, and since the light incident at 'P' on this element is reflected by the sphere along PM, so, the change in momentum of the photon beam incident on the element is directly along PN (or OP) and is given by

$$dp = 2\left(\frac{dE}{c}\right)\cos\theta = \frac{2}{c}(Idt)\left(dA\cos^2\theta\right)$$

The force (F) on dA due to the light falling on it is directed along PQ, given by

$$\boxed{F = \frac{dp}{dt} = 2\left(\frac{IdA}{c}\right)\cos^2\theta}$$

It is observed that due to symmetry, the net force on the ring element as well as the sphere is directed along with the line XO. The component of this force

(directed along with the line XO) acting on the element of area dA is:-

$$dF = F\cos\theta = \left[2\left(\frac{I dA}{c}\right)\cos^2\theta\right]\cos\theta$$

$$\Rightarrow \quad dF = F\cos\theta = \frac{2I dA}{c}\cos^3\theta$$

$$\Rightarrow \quad dF = \left(\frac{2I}{c}\right)\left(2\pi R^2 \sin\theta d\theta\right)\cos^3\theta$$

The force on the entire sphere is

$$F = \int dF$$

$$\Rightarrow \quad F = \int_0^{\pi/2}\left(\frac{2I}{c}\right)\left(2\pi R^2 \sin\theta d\theta\right)\cos^3\theta$$

$$\Rightarrow \quad F = \left(\frac{4\pi R^2 I}{c}\right)\int_0^{\pi/2}\cos^3\theta \sin\theta d\theta$$

Put $\cos\theta = t$

$$-\sin\theta d\theta = dt \Rightarrow (\sin\theta d\theta) = (-dt)$$

$$F = \frac{4\pi R^2 I}{c}\int (t)^3(-dt)$$

$$= -\frac{4\pi R^2 I}{c}\int t^3 dt$$

$$= -\frac{4\pi R^2 I}{c}\left(\frac{t^4}{4}\right)$$

$$= -\frac{4\pi R^2 I}{c}\left(\frac{\cos^4 t}{4}\right)_0^{\pi/2}$$

$$= -\frac{4\pi R^2 I}{c\times 4}\times\left[\cos^4 \pi/2 - \cos^4 0\right]$$

$$= -\frac{\pi R^2 I}{c}[0-1]$$

$$F = \frac{\pi R^2 I}{c}$$

$$F = \frac{I}{c}\times\left(\pi R^2\right)$$

$$\boxed{F = \frac{I}{c} \times (\text{Projected Area})}$$

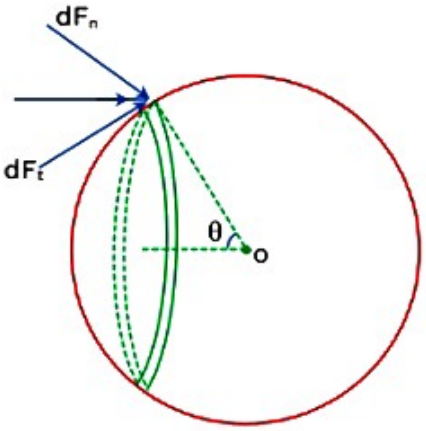
However, when the sphere is perfectly absorbing, then the normal force (dF_n) and the tangential force (dF_t) on the infinitesimal element of the area $dA = 2\pi R^2 \sin\theta d\theta$ are given by

$$dF_n = \left(\frac{I}{c}\cos^2\theta\right) dA$$

and

$$dF_t = \left(\frac{I}{c}\sin\theta\cos\theta\right) dA$$

The net force F on the perfectly absorbing sphere acts along the direction of an incident light beam, as shown in the figure.



$$\Rightarrow \quad F = \int dF = \int_0^{\pi/2} \left(dF_n \cos\theta + dF_t \sin\theta\right)$$

$$\Rightarrow \quad F = \frac{I}{c}\int_0^{\pi/2} \left(\cos^3\theta + \sin^2\theta\cos\theta\right)\left(2\pi R^2 \sin\theta d\theta\right)$$

$$= \frac{I}{c}\int_0^{\pi/2} (\cos\theta)\left[\cos^2\theta + \sin^2\theta\right]\left(2\pi R^2\right)(\sin\theta d\theta)$$

$$F = \left(\frac{\pi R^2 I}{c}\right)\int_0^{\pi/2} (2\sin\theta\cos\theta)\, d\theta$$

$$F = \frac{\pi R^2 I}{c}\int_0^{\pi/2} \sin(2\theta) d\theta$$

$$F = \frac{\pi R^2 I}{c}\left[-\frac{\cos 2\theta}{2}\Big|_0^{\pi/2}\right]$$

$$F = -\left(\frac{\pi R^2 I}{2c}\right)[\cos\pi - \cos 0]$$

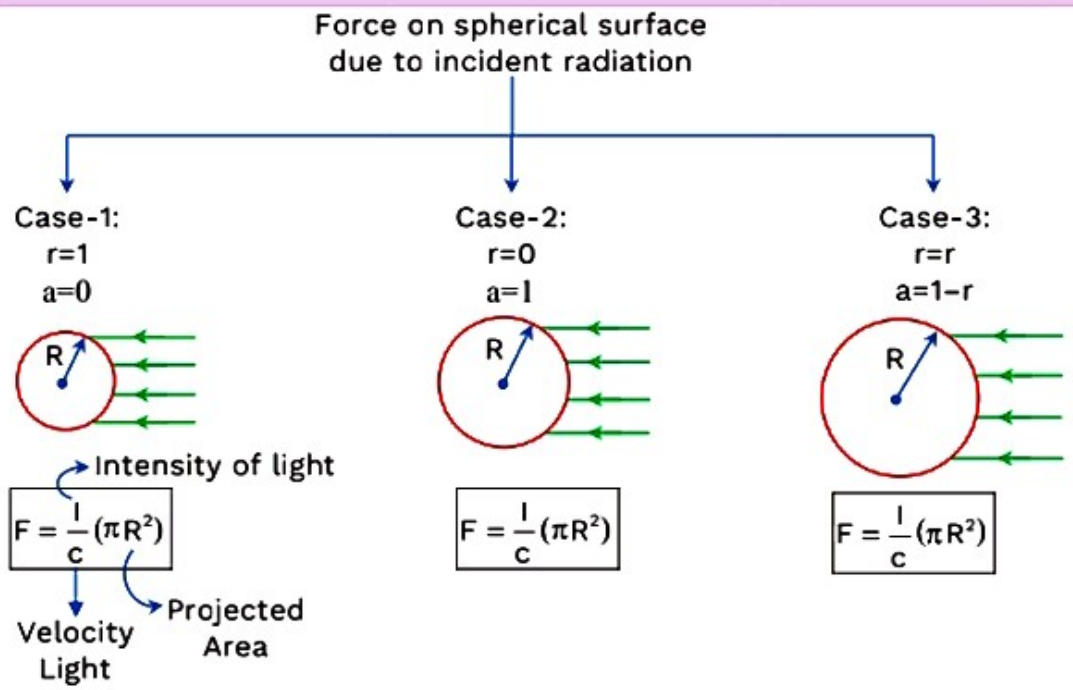
$$F = -\left(\frac{\pi R^2 I}{2c}\right)[-1-1]$$

$$F = -\left(\frac{\pi R^2 I}{2c}\right)(-2)$$

$$\boxed{F = \frac{\pi R^2 I}{c}} \Rightarrow \boxed{F = \frac{I}{c}\left(\pi R^2\right)}, \text{ where } \pi R^2 = \text{Projected area}$$

It is observed that, for a sphere placed in the path of a light beam, the force exerted on the sphere is independent of the nature of the surface of the sphere. This is because, the perfectly reflecting sphere reflects the incoming radiation in the backward sense for $\theta < \frac{\pi}{4}$ but in the forward sense for $\frac{\pi}{4} < \theta < \frac{\pi}{2}$. Due to this, the net momentum of the reflected radiation turns out to be zero and hence the net momentum exchange between the photon beam and the sphere is same in both the cases.

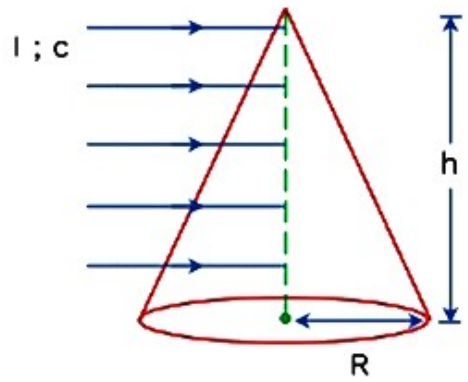
Summary



Note : In all 3 cases same force was exerted on the sphere.

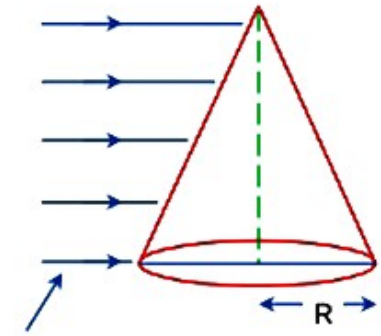
Q. **Find the force exerted on a cone having radius R and height h due to incident radiation, having Intensity I.**

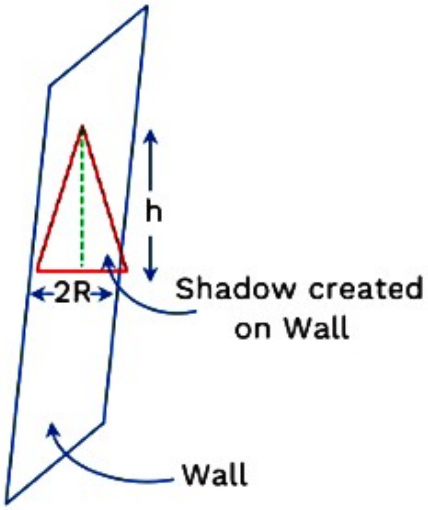
Sol.



$$\text{Force on cone} = \frac{I}{c} \times (\text{Projected Area of Cone})$$

Concept of Projected Area :-





$$\text{Projected} = \frac{1}{2} \times \text{Base} \times \text{hight}$$

$$= \frac{1}{2} \times 2R \times h$$

$$\boxed{\text{Projected Area} = Rh}$$

$$\boxed{\text{Force Exerted on Cone} = \frac{I}{c} \times R \times h}$$

Q. **In an experiment on photoelectric effect, the light of wavelength 400 nm is incident on a cesium plate at the rate of 5 W. The potential of the collector plate is made sufficiently positive with respect to the emitter so that the current reaches the saturation value. Assuming that, on average, one out of every 106 photons is able to eject a photoelectron, find the photocurrent in the circuit.**

Sol. $$E = \frac{12375}{4000} = 3.1\,eV$$

Number of photoelectrons emitted per second

$$n = \left(\frac{1}{10^6}\right)\left(\frac{5}{3.1 \times 1.6 \times 10^{-19}}\right) = 1 \times 10^{13} \text{ per second}$$

Since, $i = \frac{q}{t} = \frac{Ne}{t} = \left(\frac{N}{t}\right)e = ne$

$\Rightarrow$ $i = (ne) = 1 \times 10^{13} \times 1.6 \times 10^{-19} = 1.6 \times 10^{-6}\ A = 1.6\ \mu A$

Q. **Ultraviolet light of wavelength 2000 Å causes photoemission from a surface. The stopping potential is 2V.**
(a) Calculate the work function in eV.
(b) Calculate the maximum speed of the photoelectrons.

Sol. (a) Using Einstein relation,

$$W = \frac{hc}{\lambda} - eV_0$$

$$\Rightarrow \quad W = \frac{12400}{2000} - 2 = 4.2\,eV$$

(b) Since $\frac{1}{2}mv_{max}^2 = eV_0$.

$$\Rightarrow \quad v_{max} = \sqrt{\frac{2eV_0}{m}} = \sqrt{\frac{2(1.6\times10^{-19})(2)}{9.1\times10^{-31}}}$$

$$\Rightarrow \quad v_{max} = 8.4\times10^5\ ms^{-1}$$

Q. **When a beam of Photon of energy 10.6 eV and intensity 2 Wm^{-2} falls on a platinum surface of area $1\times10^{-1}\ m^2$ and work function 5.6 eV, 0.53% of the incident photons eject photoelectrons. Calculate the number of photoelectrons emitted per second and their minimum and maximum energies (in eV). Given that 1 eV = 1.6×10^{-19} J.**

Sol. Number of photoelectrons emitted per second

$$\frac{N}{t} = \frac{(\text{Intensity})(\text{Area})}{(\text{Energy of each photon})}\times\frac{0.53}{100}$$

$$\Rightarrow \quad \frac{N}{t} = \frac{(2)(1\times10^{-4})}{(10.6\times1.6\times10^{-19})}\times\frac{0.53}{100} = 6.25\times10^{11}s^{-1}$$

Minimum kinetic energy of photoelectrons is

$$K_{min} = 0$$

And maximum kinetic energy is,

$$K_{max} = E - W = (10.6 - 5.6)\ eV = 5\ eV$$

Q. **A monochromatic light source of frequency f illuminates a metallic surface and ejects photoelectrons. The photoelectrons of maximum energy are just able to ionize the hydrogen atoms in the ground state. When the entire experiment is repeated with incident radiation of frequency $\frac{5}{6}f$, the photoelectrons so emitted are able to excite the hydrogen atom beam, which then emits radiation of wavelength 1215 Å. (Take hc = 12375 eV–Å)**

(a) Calculate is the frequency of incident radiation.

(b) Calculate the work function of the metal.

Sol. (a) Using Einstein's equation of photoelectric effect i.e.

$$K_{max} = hf - W$$

Where $K_{max} = 13.6$ eV

$$\Rightarrow \quad hf - W = 13.6 \text{ eV} \qquad \ldots(1)$$

So, when the experiment is repeated, then

$$h\left(\frac{5}{6}f\right) - W = \frac{12375}{1215} = 10.2\,\text{eV} \qquad \ldots(2)$$

Solving equations (1) and (2), we get

$$\frac{hf}{6} = 3.4\,\text{eV}$$

$$\Rightarrow \quad f = \frac{(6)(3.4)(1.6\times10^{-19})}{(6.63\times10^{-34})} = 4.92\times10^{15}\,\text{Hz}$$

(b) From equation (1), we have

$$W = hf - 13.6$$

$$\Rightarrow \quad W = 6(3.4) - 13.6$$

$$\Rightarrow \quad W = 6.8 \text{ eV}$$

Q. **Given that the wavelength of the incident radiation is increased from 3000 Å to 3010 Å find the corresponding change in the stopping potential V.**

Sol. According to Einstein's Photo-electric equation, we have

$$eV_1 = E_1 - W \qquad \ldots(1)$$

$$eV_2 = E_2 - W \qquad \ldots(2)$$

Subtracting (2) from (1), we get

$$e(V_1 - V_2) = (E_1 - E_2)$$

$$\Rightarrow \quad V_1 - V_2 = \frac{hc}{e}\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)$$

$$\Rightarrow \quad V_1 - V_2 = \frac{6.6\times10^{-34}\times3\times10^{8}}{1.6\times10^{-19}\times10^{-10}}\left(\frac{1}{3000} - \frac{1}{3010}\right) = 0.012\,\text{V}$$

Q. **Light of wavelength 180 nm ejects photoelectrons from a plate of a metal whose work function is 2 eV. If a uniform magnetic field of 50 μT is applied parallel to the plate, what would be the radius of the path followed by electrons ejected normally from the plate with maximum energy.**

Sol. $\lambda = 180 \text{ nm} = 1800 \text{ Å}$

$$\Rightarrow \quad E = \frac{12375}{1800} = 6.875 \text{ eV}$$

$$K_{max} = E - W = 4.875 \text{ eV}$$

Since, $r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$

Substituting the values, we get

$$r = \frac{\sqrt{2 \times 4.875 \times 1.6 \times 10^{-19} \times 9.1 \times 10^{-31}}}{5 \times 10^{-5} \times 1.6 \times 10^{-19}} = 0.15 \text{ m} = 15 \text{ cm}$$

Q. **There is photoelectric effect set-up in which a point source of light of power 3.2×10^{-3} W emits monoenergetic photons of energy 5 eV. The source of light is located at a distance of 0.8 m from the centre of a stationary metallic sphere. The work function of the sphere and radius are 3 eV and 8×10^{-3} m, respectively given that the efficiency of photo-electrons emission is one for every 10^6 incident photons. Assumption: the sphere is isolated, and electrons are instantly swept away after emission.**

(a) Find the number of photoelectrons emitted per second.

(b) Find the ratio of the wavelength of incident light to the de Broglie wavelength of the fastest photoelectrons emitted.

(c) It is observed that the photoelectron emission stops at a certain time t after the light source is switched on. Why?

(d) Evaluate the time t.

Sol. (a) Number of photoelectrons emitted per second is

$$n = \left(\frac{1}{10^6}\right)\left(\frac{P}{E}\right)\left(\frac{\pi r^2}{4\pi R^2}\right)$$

$$\Rightarrow \quad n = \left(\frac{1}{10^6}\right)\left(\frac{3.2 \times 10^{-3}}{5 \times 1.6 \times 10^{-19}}\right)\left(\frac{(\pi \times 8 \times 10^{-3} \times 8 \times 10^{-3})}{4\pi \times 0.8 \times 0.8}\right)$$

$\Rightarrow \quad n = 10^5 \text{ sec}^{-1}$

(b) $K_{max} = E - W = 2eV$

Since, for an electron, we have $\lambda = \dfrac{h}{\sqrt{2m_e(kE)}} = \sqrt{\dfrac{150}{KE(\text{in eV})}}$ Å

$\Rightarrow \quad \lambda_2 = \sqrt{\dfrac{150}{2}} = 8.66$ Å

Further, the wavelength of the incident photon is

$$\lambda_1 = \frac{12375}{5} = 2475 \text{ Å}$$

$\Rightarrow \quad \dfrac{\lambda_1}{\lambda_2} \approx 284$

(c) Photoemission will stop when the potential on the sphere becomes equal to the stopping potential.

(d) $K_{max} = 2 \text{ eV}_0$. Therefore, the stopping potential V_0 is 2 volt. Let t be the desired time. Then

$$V_0 = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} = \frac{Ne}{4\pi\varepsilon_0 r} = \frac{(nt)e}{4\pi\varepsilon_0 r}$$

$$\Rightarrow \quad t = \frac{V_0 r}{\left(\dfrac{1}{4\pi\varepsilon_0}\right)(ne)} = \frac{2 \times 8 \times 10^{-3}}{9 \times 10^9 \times 10^5 \times 1.6 \times 10^{-19}} = 111\text{s}$$

Q. **Light is incident on the cathode, and the stopping voltages are measured for the light of two different wavelengths. From the data given below, determine the work functions of the metal of the cathode in eV.**

Wavelength (Å)	Stopping voltage (volt)
4000	**1.3**
4500	**0.9**

Sol. As $K_{max} = hf - W_0$

or $\quad eV_0 = \dfrac{hc}{\lambda} - W$

or $\quad V_0 = \dfrac{hc}{e\lambda} - \dfrac{W_0}{e}$

$\therefore \quad \Delta V_0 = (V_0)_2 - (V_0)_1$

$$= \left[\frac{hc}{e\lambda_2} - \frac{W_0}{e}\right] - \left[\frac{hc}{e\lambda_1} - \frac{W_0}{e}\right] = \frac{hc}{e}\left[\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right]$$

$$\Delta V_0 = \frac{hc}{e}\left[\frac{\lambda_1 - \lambda_2}{\lambda_1\lambda_2}\right]$$

$$\therefore \quad \frac{hc}{e} = \frac{\Delta V_0 . \lambda_1\lambda_2}{\lambda_1 - \lambda_2}$$

$$= \frac{(1.3 - 0.9) \times 4000 \times 10^{-10} \times 4500 \times 10^{-10}}{500 \times 10^{-10}}$$

$$= 1.44 \times 10^{-6} \text{ Vm}$$

Also, $$V_0 = \frac{hc}{e\lambda} - \frac{W_0}{e}$$

$$\therefore \quad \frac{W_0}{e} = \frac{hc}{e\lambda} - V_0 = \frac{1.44 \times 10^{-6}}{4000 \times 10^{-10}} - 1.3 = 2.3 \text{ V}$$

$$\Rightarrow \quad W_0 = 2.3 \text{ eV}$$

Q. **The photoelectric threshold of silver is λ = 3800 Å. Ultraviolet light of λ = 2600 Å is incident on a silver surface. Calculate:**

(a) The value of work function in joule and in eV.

(b) Maximum kinetic energy of the emitted photoelectrons.

(c) The maximum velocity of the photoelectrons.

(Mass of the electrons = 9.11×10^{-31} kg)

Sol. (a) $\lambda_0 = 3800$ Å

$$W_0 = hf_0 = h\frac{c}{\lambda_0} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{3800 \times 10^{-10}}$$

$$= 5.23 \times 10^{-19} \text{ J} = 3.27 \text{ eV}$$

(b) Incident wavelength, λ = 2600 Å

Therefore, f = incident frequency $= \frac{3 \times 10^8}{2600 \times 10^{-10}}$ Hz

$$hf = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2600 \times 10^{-10}} = 7.65 \times 10^{-19} \text{J} = 4.78 \text{ eV}$$

then, $K_{max} = hf - W_0 = 4.78 \text{ eV} - 3.27 \text{ eV} = 1.51 \text{ eV} = 2.42 \times 10^{-19}$ J

(c) $K_{max} = \frac{1}{2}mv_{max}^2$

$$\Rightarrow \quad v_{max} = \sqrt{\frac{2K_{max}}{m}} = \sqrt{\frac{2 \times 2.42 \times 10^{-19}}{9.11 \times 10^{-31}}}$$
$$= 0.7289 \times 10^{6}\ ms^{-1}$$

Q. **A light ray of wavelength 4950 Å is incident on a surface, and a photocurrent appears, which vanishes if a retarding potential greater than 0.6 Volt is applied across the photo electric tube. When light from a different source is used, it is found that the critical retarding potential is changed to 1.1 V. Find the work function of the emitting surface and the wavelength of the second source.**

Sol. We know that $hf_1 = \phi + \frac{1}{2}mv_1^2$ and $\frac{1}{2}mv_1^2 = eV_1$

$$\phi = hf_1 - eV - eV_1 = \frac{hc}{\lambda_1} - eV_1$$

$$= \frac{(6.6 \times 10^{-34}) \times (3 \times 10^{8})}{4950 \times 10^{-10}} - (1.6 \times 10^{-19})(0.6)$$

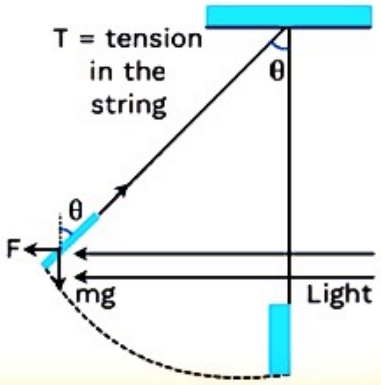
$$= 3.04 \times 10^{-19}\ V$$

$$hf_2 = \phi + eV_2$$

$$\frac{hc}{\lambda_2} = 3.04 \times 10^{-19} + (1.6 \times 10^{-19}) \times 1.1 = 4.8 \times 10^{-19}$$

$$\Rightarrow \quad \lambda_2 = \frac{(6.6 \times 10^{-34}) \times (3 \times 10^{8})}{4.8 \times 10^{-19}} = 4125\ Å$$

Q. **There is a plain strip of mass m suspended from fixed support through a string. A continuous beam of monochromatic light is incident horizontally on the strip and is completely absorbed. The energy falling on the strip per unit time is W. Find the deflection of string from the vertical if the strip stays in equilibrium.**



Sol. The radiation incident on the strip applies force on it; because of this force, the strip is deflected from its original position.

Force exerted by the photons of light = (number of incident photons per second) × (change in momentum of each photon)

Let n be the number of incident photons per second.
The momentum of photon incident on the strip per second,

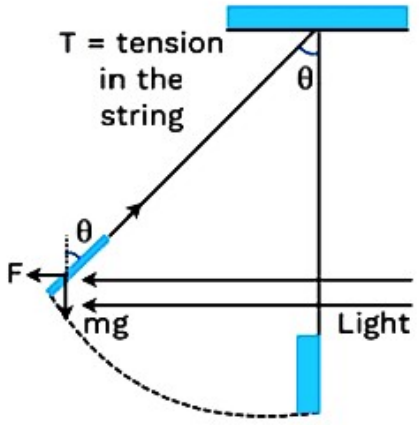
$$P_{incident} = \frac{W}{c}$$

As the photon is completely absorbed, hence its final momentum should be zero.

The change in momentum of each photon per second, $\left|\frac{\Delta p}{\Delta t}\right| = \frac{W}{c}$

Force acting on the strip due to radiation, $F = \frac{W}{c}$

Suppose the angle made by the string from the vertical direction is θ. Now At the equilibrium condition of the strip, we can write the following equation,



For the, horizontal direction; $T \sin\theta = F$.
And, for the vertical direction; $T \cos\theta = mg$.

Dividing the above equations, we get

$$\tan\theta = \frac{F}{mg} = \frac{W/c}{mg} = \frac{W}{cmg}$$

$$\tan\theta = \frac{W}{cmg}$$

$$\theta = \tan^{-1}\left(\frac{W}{cmg}\right).$$

Matter Waves

Introduction:

All matter exhibits wave-like behavior. For example, a beam of electrons can be diffracted, just like a beam of light. In most cases, however, the wavelength is too small to have a practical impact on day-to-day activities. Hence in our day-to-day lives, with an object the size of tennis balls and people, matter-wave is not relevant.

The concept that matters behaves like a wave was proposed by French physicist Louis de Broglie in 1924. It is also referred to as the de Broglie hypothesis. Matter waves are referred to as de Broglie waves.

De-Broglie Hypothesis:-

De-Broglie imagined that as light possesses both wave and particle nature, similarly matter must also possess both natures, particle as well as wave. De-Broglie imagined that despite the particle nature of matter, waves must also be associated with material particles, known as matter waves.

De-Broglie Wave Length Associated with Moving Particles :-

If a particle of mass 'm' is moving with velocity 'v', then

Kinetic Energy of the Particle :-

$$\text{Kinetic Energy} = \frac{1}{2}mv^2$$

$$= \frac{1}{2m} \times m^2v^2$$

$$\text{Kinetic Energy} = \frac{p^2}{2m}$$

$$\boxed{p = \sqrt{2m \times (K.E)}}$$

- The wavelength associated with the particle is:-

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2m(K.E)}}$$

$$\boxed{\lambda = \frac{h}{\sqrt{2m(K.E)}}}$$

Summary

De-Broglie Wavelength

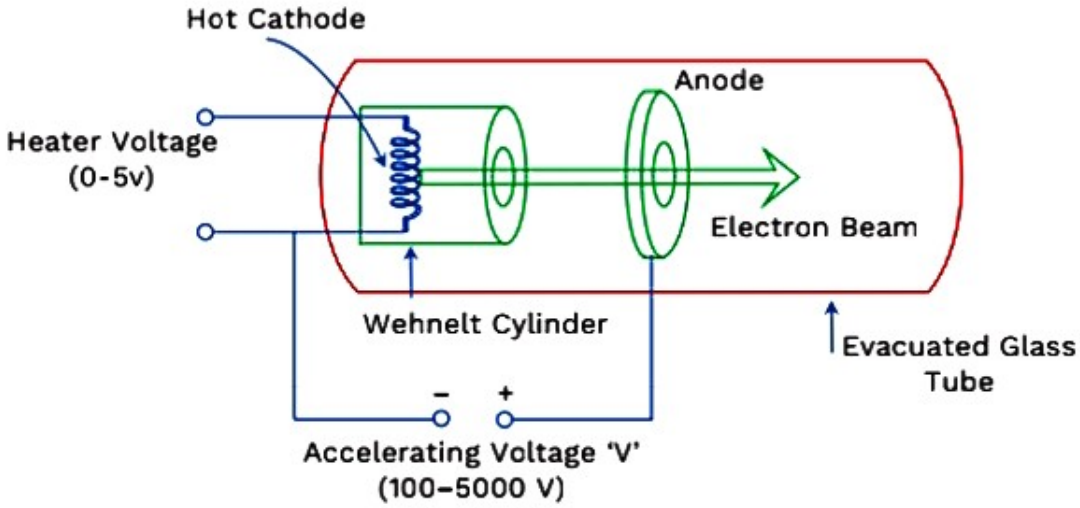
$\lambda = \frac{h}{p}$; if p is given

$\lambda = \frac{h}{mv}$; if mass and v are given.

$\lambda = \frac{h}{\sqrt{2m(KE)}}$; if Kinetic Energy is given.

De Broglie Wave Length Associated with the Charged Particles:-

Case -1: De Broglie Wavelength associated with electron: -



Consider an electron (Mass m_e and charge 'e') accelerated from rest through a potential 'V'. The Kinetic Energy (K.E) of the electron is equal to work done (eV) on it by the electric field.

$$\boxed{\text{Kinetic Energy} = eV} \qquad ...(1)$$

Now,

$$K.E = \frac{1}{2}m_e v^2 = \frac{p^2}{2m_e} \Rightarrow p = \sqrt{2m_e(K.E.)} \qquad ...(2)$$

The de Broglie wavelength of the electron is then

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(K.E)}} = \frac{h}{\sqrt{2m_e eV}}}$$

Now put the values of constant :-

$h = 6.634 \times 10^{-34}$ Jr

$e = 1.6 \times 10^{-19}\ C$

$m_e = 9.1 \times 10^{-31}\ kg$

$$\lambda = \frac{6.634 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times V}}$$

$$\lambda = \frac{6.634 \times 10^{-9}}{\sqrt{2 \times 9.1 \times 1.6}} \times \frac{1}{\sqrt{V}}$$

$$\lambda = \frac{1.22736 \times 10^{-9}}{\sqrt{V}}$$

$$\boxed{\lambda = \frac{12.27}{\sqrt{V}}(A°)}$$

Wave Length of electron in A°

Accelerating Potential in volt (V)

Q. **An electron is accelerated with 16 volts. Find de Broglie wavelength of the electron.**

Sol. $$\lambda = \frac{12.27A°}{\sqrt{16}} = \frac{12.27A°}{4}$$

$$\boxed{\lambda = 3.0675\ A°}$$

Case -2 : De Broglie wavelength associated with proton:-

$$\lambda = \frac{h}{\sqrt{2 \times m_p \times e \times V}}$$

$$\boxed{\lambda = \frac{0.286}{\sqrt{V}}(A°)}$$

de Broglie 'λ' in A°

Accelerating Voltage in 'Volt'

Case -3 : Associated with alpha (α) Particles :-

De Broglie Wave Length (λ)

$$\lambda = \frac{h}{\sqrt{2 \times m_\alpha \times e \times V}}$$

$$\lambda = \frac{0.101}{\sqrt{V}}$$

The wavelength of Alpha Particles

"Accelerating Voltage"

De-Broglie Wave Length Associated with Uncharged Particles :-

If an uncharged particle has thermal Kinetic Energy

$$E = \frac{3}{2}kT$$

Then it de Broglie Wavelength

$$\lambda = \frac{h}{\sqrt{2mK.E}}$$

$$\lambda = \frac{h}{\sqrt{2 \times m \times \frac{3}{2}kT}}$$

$$\lambda = \frac{h}{\sqrt{3m \times kT}}$$

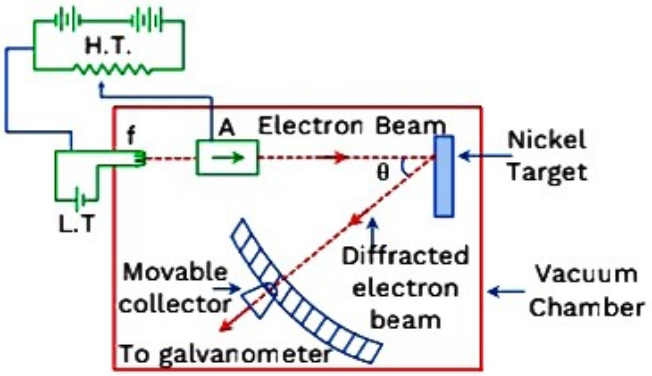
k = Boltzmann Constant.
m = Mass of Particle.
T = Temperature.

Davisson and Germer Experiment :-

"The wave nature of electron was first experimentally verified by C-J Davisson and E.H. Germer in 1927 and independently by G.P-Thomson in 1928, who observed diffraction effect with a beam of electrons scattered by a crystal.

(They Got Nobel Prize in 1937)

Experimental Arrangement :-



The experimental arrangement consists of an electron gun which comprises of a tungsten filament f, coated with barium oxide and heated by a low voltage Power Supply (L. T., or battery).

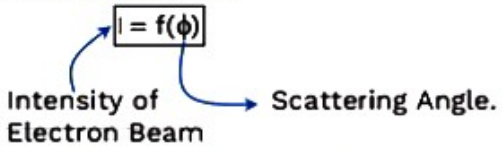
Electrons emitted by the filament are accelerated to a desired velocity by applying suitable potential/voltage from a high voltage power supply (H. T. or battery).

They are made to pass through a cylinder with fine holes along its axis, producing a finely collimated beam. The beam is made to fall on the surface of a nickel crystal. The electron is scattered in all directions by the atoms of the crystal. The intensity of the electron beam scattered in a given direction is measured by the electron detector (Collector).

The detector can be moved on a circular scale and is connected to a sensitive galvanometer, which records the current.

Measurement and Analysis:-

The intensity (I) of the scattered electron beam is measured for different values of angle of scattering (ϕ), which is the angle between the incident and the scattered electron beams. The variation of the intensity (I) of the scattered electron with the angle of scattering is obtained for different accelerating voltages.



The experiment was performed by varying the accelerating voltage from 44V to 68V. It was noticed that a strong peak appeared in the intensity (I) of the scattered electron for an accelerating voltage of 54V at a scattering angle ϕ = 50°.

The graph between scattering angle ϕ and intensity (I)

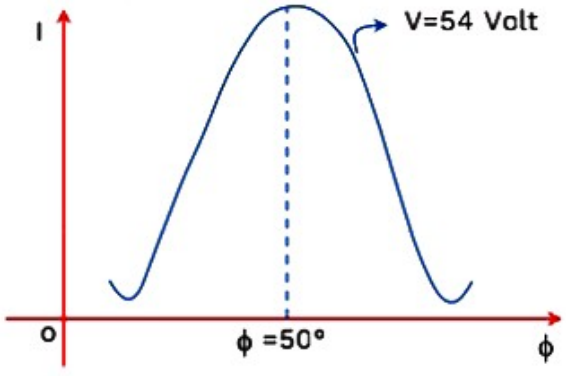
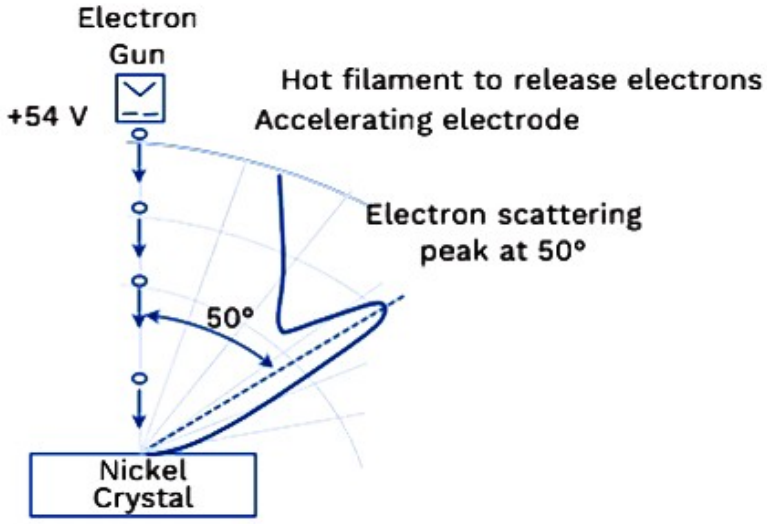
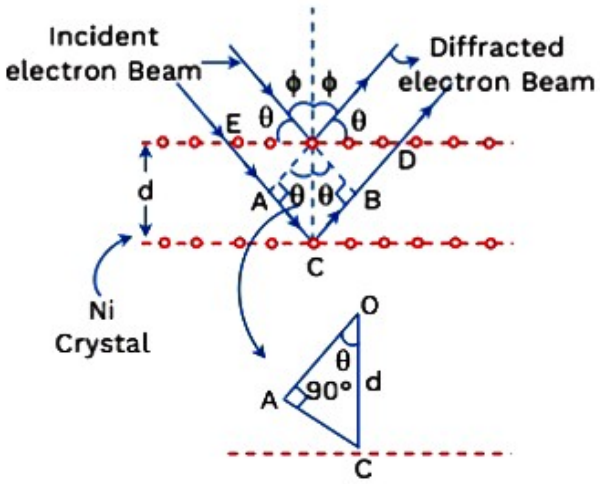


Figure (a)



Calculation of wavelength (λ) using experimental set up (Davisson – Germer experiment)



$$\sin\theta = \frac{AC}{OC}$$

$$\sin\theta = \frac{AC}{d}$$

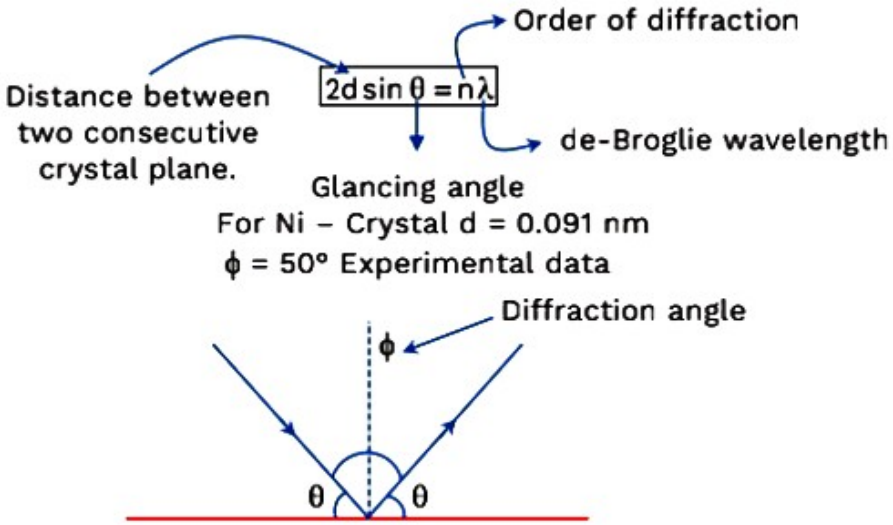
$$AC = d\sin\theta$$

So, Path difference

$$\boxed{\Delta x = 2d\sin\theta}$$

Since the beam of electron diffracted from different layers of Ni- crystal due to this reason, a path difference is created between matter waves.

By using the concept of constructive interference of waves, we can write:



$2\theta + \phi = 180°$

$\phi = 50°$

$2\theta = 180° - 50$

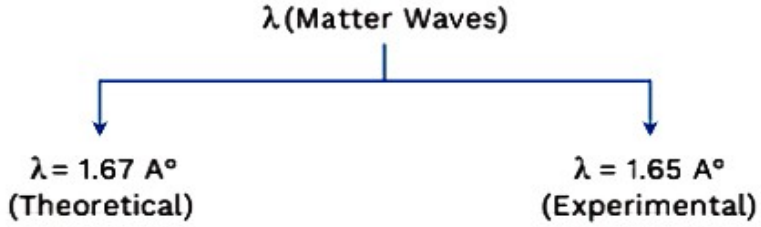
$2\theta = 130°$

$\theta = 65°$

$2 \times 0.091 \times 10^{-9} \times \sin 65° = 1 \times \lambda$

$\boxed{\lambda = 1.65\ A°}$

Wavelength calculated using experimental set up.

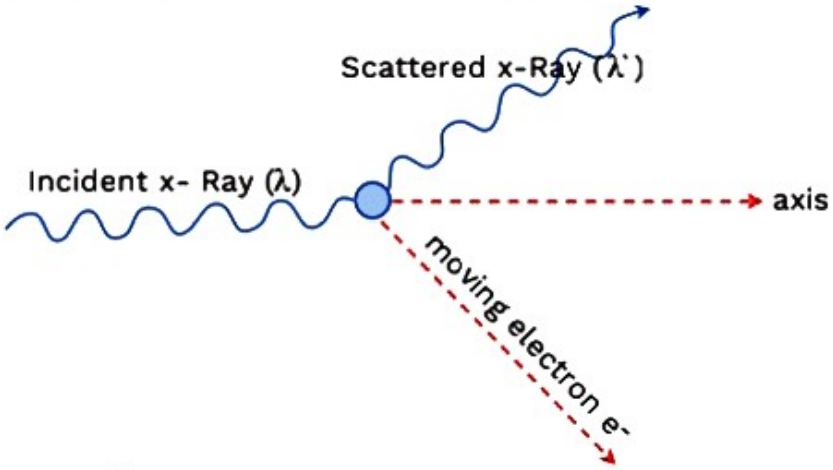


Thus, there is excellent agreement between the theoretical value and the experimentally obtained value of de-Broglie wavelength.

Davisson Germer's experiment thus strikingly confirms the wave nature of an electron and de-Broglie relations.

Compton-Effect :-

- The Compton Effect was an experiment conducted by Author H. Compton in 1923 that further confirmed the quantum theory of light. Compton directed electromagnetic waves with short-wavelength (x-Rays) at electron and found that incident waves had slightly shorter wavelength than the waves produced after the collision.



$$\lambda` = \lambda + \frac{h}{m_e c}(1 - \cos\phi)$$

The shift in wavelength is given by

$$\Delta\lambda = \lambda` - \lambda = \frac{h}{m_e c}(1 - \cos\phi)$$

This is known as the Compton shift.

What Wave Theory Predicts :-

- The wave theory predicts that no wavelength change should take place.
- The incoming E.M. wave causes the electron to oscillate with the same frequency as the wave. Therefore, the oscillating electron should remit the EM wave with the same frequency.

Confirmation of Quantum Theory :-

Incoming Photon collides with an electron and transfers some of the energy to the electron. The scattered Photon now has less energy than before and so decreases its frequency (E = hf) and increases in wavelength $\left(\lambda = \frac{c}{f}\right)$.

Compton wavelength $(\lambda_c) = \frac{\lambda}{m_e c} = 2.43 \times 10^{-3} nm$.

Important - "Compton Effect confirmed the 'quantum theory of light'.

Q. **Calculate the de-Broglie wavelengths for electrons and protons if their speed is 10^5 m/s.**

Sol. We are given that $v = 10^5$ m/s.

(i) Mass of an electron, $m = 9.1 \times 10^{-31}$ kg

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(9.1 \times 10^{-31})10^5}$$

or $\lambda = 72.8 \times 10^{-10} m = 72.8$ Å

(ii) Mass of proton, $m = 1.67 \times 10^{27}$ kg

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(1.67 \times 10^{-27})10^5}$$

or $\lambda = 0.03968 \times 10^{-10}$ m $= 0.03968$ Å

Q. **Calculate the de-Broglie wavelength associated with an α-particle accelerated through a potential difference of 200 V.**
Given $m_p = 1.67 \times 10^{-27}$ kg.

Sol. Mass of α-particle, $m = 4m_p = 4 \times 1.67 \times 10^{-27}$ kg

Charge on α-particle = 2e

If the α-particle acquires velocity v, then $qV = \frac{1}{2}mv^2$

or $2eV = \frac{1}{2m}m^2v^2$

$\therefore \quad mv = \sqrt{4meV}$

$\Rightarrow \quad \lambda = \frac{h}{mv} = \frac{h}{\sqrt{4meV}}$

$$\Rightarrow \quad \lambda = \frac{6.6 \times 10^{-34}}{\sqrt{4 \times 4 \times 1.67 \times 10^{-27} \times 1.6 \times 10^{-19} \times 200}} \text{m}$$

$$= 0.07138 \times 10^{-11} \text{ m} = 0.007138 \text{ Å}$$

Q. **(a) The de-Broglie wavelength is 1.40×10^{-10} m associated with a neutron; find the kinetic energy of neutron?**

(b) Also, find the de-Broglie wavelength of a neutron, in thermal equilibrium with matter, having an average kinetic energy of (3/2) kT at 300 K. Given that mass of a neutron.
$= 1.66 \times 10^{-27}$ kg and k = 1.38×10^{-23} J kg^{-1}.

Sol. **(a)** Here $\lambda = 1.40 \times 10^{-10}$ m; $m = 1.66 \times 10^{-27}$ kg

The energy of neutron is given by $E = \frac{h^2}{2m\lambda^2}$

$$= \frac{(6.62 \times 10^{-34})^2}{2 \times 1.66 \times 10^{-27} \times (1.40 \times 10^{-10})^2} = 6.735 \times 10^{-21} \text{J}$$

(b) Here $k = 1.38 \times 10^{-23}$ J kg^{-1} and T = 300 K

Therefore, energy of neutron

$$E = \frac{3}{2}kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300 = 6.21 \times 10^{-21} \text{J}$$

$$\text{Now,} \quad \lambda = \frac{h}{\sqrt{2mE}} = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 1.66 \times 10^{-27} \times 6.21 \times 10^{-21}}}$$

$$= 1.46 \times 10^{-10} \text{ m} = 1.46 \text{ Å}$$

Q. **The wavelength of light from the spectral emission line of sodium is 589 nm. Find the kinetic energy at which (a) an electron and (b) a neutron would have the same de Broglie wavelength. Given that mass of neutron = 1.66×10^{-27} kg.**

Sol. Here, $\lambda = 589 \text{ nm} = 589 \times 10^{-9}$ m

The de Broglie wavelength of a particle having energy E is given by

$$\lambda = \frac{h}{\sqrt{2mE}} \quad \text{or} \quad E = \frac{h^2}{2m\lambda^2}$$

(a) For electron: Here, $\lambda = 589\ nm = 589 \times 10^{-9}\ m$ and $m = 9.1 \times 10^{-31}\ kg$

$\therefore$

$$E = \frac{h^2}{2m\lambda^2} = \frac{(6.62 \times 10^{-34})^2}{2 \times 9.1 \times 10^{-31} \times (589 \times 10^{-9})^2}$$

$$= 6.94 \times 10^{-25}\ J$$

(b) For neutron: Here $\lambda = 589\ nm = 589 \times 10^{-9}\ m$

$m = 1.66 \times 10^{-27}\ kg$

$$\therefore \quad E = \frac{h^2}{2m\lambda^2} = \frac{(6.62 \times 10^{-34})^2}{2 \times 1.66 \times 10^{-27} \times (589 \times 10^{-9})^2}$$

$$= 3.81 \times 10^{-28}\ J$$

Q. **Obtain the de-Broglie wavelength associated with thermal neutrons at room temperature (27°C). Hence explain why a fast neutron beam needs to be thermalized with the environment before it can be used for neutron diffraction experiments.**

Sol. The average kinetic energy of a neutron at absolute temperature T is

$$\frac{1}{2}mv^2 = \frac{3}{2}k_B T$$

$$\Rightarrow \quad \frac{p^2}{2m} = \frac{3}{2}k_B T \qquad \{\because p = mv\}$$

$$\Rightarrow \quad p = \sqrt{3mk_B T}$$

de-Broglie wavelength is given by

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{3mk_B T}}$$

Given $m_n = 1.675 \times 10^{-27}\ kg$, $k_B = 1.38 \times 10^{-23}\ J\ mol^{-1}\ K^{-1}$

$T = 27 + 273 = 300\ K$, $h = 6.63 \times 10^{-34}\ Js$

$$\Rightarrow \quad \lambda = \frac{6.63 \times 10^{-34}}{\sqrt{3 \times 1.675 \times 10^{-27} \times 1.38 \times 10^{-23} \times 300}}$$

$$\Rightarrow \quad \lambda = \frac{6.63 \times 10^{-10}}{4.56} \text{m} = 1.45 \times 10^{-10} \text{ m}$$

$$\Rightarrow \quad \lambda = 1.45 \text{ Å}$$

As this wavelength is comparable to interatomic spacing (~1 Å) in a crystal, so thermal neutron can be used for diffraction experiments. A high energy neutron beam should be first thermalized before using it for diffraction.

Q. **Sunlight intensity on the surface of the earth is 1400 Wm^{-2}. Assuming the mean wavelength of sunlight to be 6000 Å, calculate.**
(a) The photon flux arriving at 1 m^2 area on earth perpendicular to light radiations, and
(b) The number of photons emitted from the sun per second assuming the average radius of Earth's orbit is 1.49 × 10^{11} m. (Take hc = 12400 eV–Å)

Sol. **(a)** Energy of a photon

$$E = \frac{hc}{\lambda} = \frac{12400}{6000} = 2.06 \text{ eV} = 3.3 \times 10^{-19} \text{ J}$$

$$\text{Photon flux} = \frac{IA}{E} = \frac{(1400)(1)}{3.3 \times 10^{-19}} = 4.22 \times 10^{21}$$

(b) $n = \frac{P}{E} = \frac{I(4\pi R^2)}{E}$

$$\Rightarrow \quad n = \frac{(1400)(4\pi)(1.49 \times 10^{11})^2}{3.3 \times 10^{-19}} = 1.18 \times 10^{45}.$$